PROJECT REPORT

EV Market Performance & Insights

Electric vehicle sales, battery technology and driving range analytics

Project	EV Market Performance & Insights
Prepared by	Dhruv Shrivastava
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Project type	Student analytics project
Dataset	200,000 vehicle records; 37 source fields
Deliverables	Website dashboard, Tableau story, prepared data and project report

A guided analysis of recorded electric vehicle market indicators through interactive visualizations.

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Interpretation note. Values describe the supplied dataset and should not be treated as audited market totals or forecasts. Annual sales units and customer ratings include missing entries; each measure excludes blank values for that calculation.

1. INTRODUCTION

1.1 Project Overview

EV Market Performance & Insights is a student analytics project that presents recorded electric vehicle indicators through a browser dashboard and a Tableau story. Users can compare sales by manufacturer and category, inspect battery technology mix, and review driving range by vehicle category.

1.2 Purpose

• Present selected vehicle market measures in an accessible, filterable dashboard.

• Guide viewers through a short sequence of market comparisons.

• Package the website, prepared data, Tableau workbook and documentation for review.

1.3 Scope

The project covers the supplied 200,000-row training dataset and ten fields selected for the dashboard. It is descriptive and exploratory; it does not establish market representativeness, causal relationships or forecast accuracy.

2. IDEATION PHASE

2.1 Problem Statement

Vehicle sales, battery types and range can be difficult to compare across a large flat dataset. The project groups these measures into summary cards, charts, filters and a guided story so a reviewer can move from an overview to focused comparisons.

2.2 User Perspective

User need	Project response
Understand the dashboard quickly	Overview KPIs and concise chart titles
Compare market segments	Category, manufacturer, country and model-year filters
Explore technology choice	Battery type distribution view
Review a guided narrative	Five sequential Tableau story points

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User action	Expected experience
Discover	Open the project site	Understand purpose and available views
Explore	Choose filters	Focus the comparisons
Interpret	Review charts and story	Compare recorded patterns
Share	Open project resources	Access source, workbook and documentation

3.2 Solution Requirements

• Responsive web page with KPI cards, charts and filter controls.

• Views for category, year, manufacturers, market share, battery mix, range and country.

• Prepared dataset with documented fields and missing-value handling.

• Tableau workbook containing the five-point story sequence.

• Project documentation with links and clearly stated completion status.

3.3 Conceptual Data Flow

Input	Processing	Output
Supplied EV training data	Select ten fields; trim text; preserve blank values	Compressed prepared extract and Tableau data source
Prepared records	Apply dashboard filters and aggregate selected metrics	KPIs, charts and guided story

3.4 Technology Stack

Technology	Role
HTML, CSS, JavaScript	Static browser dashboard and interactions
Gzip CSV	Compact prepared data extract loaded in browser
Tableau Public workbook	Interactive analytical views and story points
GitHub	Project source and deliverable repository

4. PROJECT DESIGN

4.1 Problem-Solution Fit

Summary cards support quick scanning, filters support focused exploration, and the story sequence moves from overview to manufacturer performance, market share, battery technology and range.

4.2 Proposed Solution

• A standalone static website with five filters and interactive market views.

• A Tableau workbook with five story points and descriptive captions.

• A prepared extract retaining all 200,000 rows and the ten dashboard fields.

• A project report and source notes describing scope, data decisions and pending delivery steps.

4.3 Solution Architecture

Presentation layer	Analytics layer	Publishing layer
Responsive HTML/CSS/JavaScript site	Prepared EV records; filtered KPIs and charts; Tableau views	GitHub repository; GitHub Pages and Tableau Public links when published

5. PROJECT PLANNING & SCHEDULING

Phase	Work item	Deliverable / evidence
Define	Set scope and comparison questions	Project objectives
Prepare	Inspect fields, blanks and duplicate IDs	Prepared dataset and notes
Build	Create web dashboard and views	HTML, CSS and JavaScript
Narrative	Arrange Tableau story sequence	Five captions and views in packaged workbook
Document	Describe data, site, story and status	This project report and docs/
Publish	Deploy site, publish workbook and record demo	External links; still to be completed/verified

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Test Plan

Test area	Check	Expected result
Website load	Open index.html through HTTP/HTTPS	Dashboard and data appear
Filters	Change category, maker, battery, country or year	Displayed measures respond to selection
Reset	Activate reset control	Default filter selections return
Story sequence	Navigate all five points	Each caption and relevant view appears
Responsive layout	Check desktop and narrow viewport	Content remains readable
External links	Open published resources when available	Correct resource opens

6.2 Performance Testing

No measured browser timings or stress-test results are available yet. Record compressed data load, filter response, chart update and narrow-screen behavior in docs/performance-testing.md using a named browser/device and repeatable method. These are planned checks, not benchmark claims.

7. RESULTS

7.1 Project Outputs

• A static website dashboard with filters and market comparison views.

• A prepared gzip CSV extract containing 200,000 rows and ten selected fields.

• A packaged Tableau workbook with the following story sequence:

1. EV Market Overview — Sales, Battery Technology & Driving Range
2. Manufacturer Sales Performance
3. Manufacturer Market Share — Comparing Annual Sales
4. Battery Technology Distribution
5. Driving Range by Vehicle Category

7.2 Data Preparation Summary

The source contains 200,000 records and 37 fields. Ten dashboard fields were retained, headers were made readable, and surrounding whitespace was trimmed. Blank values were preserved. There are 2,028 blank annual sales values and 3,977 blank ratings. There are 199,999 distinct vehicle IDs, so one identifier repeats; both rows remain in the extract.

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

• Combines sales, battery mix and range comparisons in a single exploratory interface.

• Uses a prepared compressed extract and a static website without a build step.

• Provides a clear five-part Tableau narrative.

• Documents missing-value handling and avoids claiming missing metrics were imputed.

8.2 Limitations

• The supplied dataset’s representativeness and metric definitions have not been independently validated.

• Annual sales values may contain extreme values; inspect these before making conclusions.

• The dashboard counts source rows; it does not equate row count to distinct vehicles.

• GitHub Pages, external Tableau publication, measured performance results, and the demo video remain pending or require verification.

9. CONCLUSION

The project organizes a large EV dataset into a website dashboard and a guided Tableau story. It gives reviewers a practical way to compare recorded sales, battery technology and driving range while preserving clear caveats about data quality and interpretation. Completing the external publishing and demonstration steps will make the project easier to review.

10. FUTURE SCOPE

Next steps include validating metric outliers, recording browser performance, enabling GitHub Pages, publishing verified Tableau links, recording the 5–7 minute demo, and improving accessibility and link checks.

11. APPENDIX

11.1 Project Links

Resource	Clickable project link / status
GitHub repository	Open repository
Website source	Open index.html
Tableau workbook	Download packaged workbook
Project report	Open report in repository
Prepared dataset	Open cleaned data folder
Story notes	Open five-point story
Live website / demo video	Pending: add verified Pages and Google Drive URLs

11.2 Data Notes

The original ev_market_train.csv remains local; the repository holds only the prepared extract. The source has 2,028 blank sales values, 3,977 blank ratings and one repeated vehicle ID. See data/raw/README.md and docs/data-preparation.md for details.

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